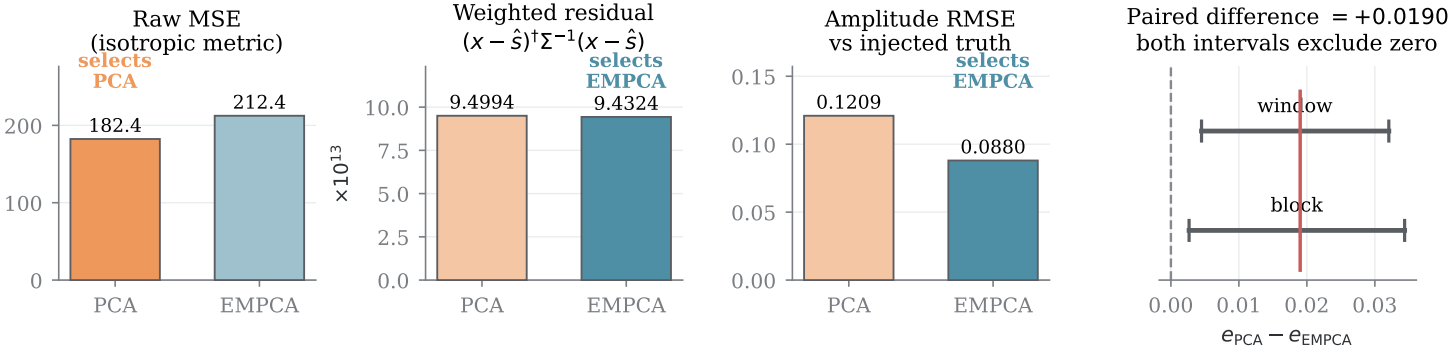
Raw MSE
(isotropic metric)
selects
PCA
182.4
212.4
200
100
0
PCA
EMPCA
Weighted residual
$(x-\hat{s})^{\dagger}\Sigma^{-1}(x-\hat{s})$
selects
EMPCA
9.4994
9.4324
$\times 10^{13}$
10.0
7.5
5.0
2.5
0.0
PCA
EMPCA
Amplitude RMSE
vs injected truth
selects
EMPCA
0.1209
0.0880
0.15
0.10
0.05
0.00
PCA
EMPCA
Paired difference = +0.0190
both intervals exclude zero
window
block
0.00
0.01
0.02
0.03
$e_{\mathrm{PCA}} - e_{\mathrm{EMPCA}}$